

LAB REPORT

Lab Title	Bitwise Operation with Command-Line Argument	Lab No.	13
Student Name	Li Rongyao	Student ID	2024103975
Date	2025/5/26		
Lab description/objectives: 1. Demonstrate understanding of the bitwise operation and the usage of command line. 2. Implement the bitwise operation with command-line argument as defined in the lab tasks/requirements. 3. Analyze the error handling and the validity of argument.			
Source code: <pre>#include <stdio.h> #include <string.h> void And(int num1, int num2); void Or(int num1, int num2); void Xor(int num1, int num2); int main(int argc, char* argv[]){ if (argc != 4){ printf("Invalid argument count.\n"); return 1; } int num1, num2; if (!sscanf(argv[1], "%d", &num1) !sscanf(argv[3], "%d", &num2)){ printf("Invalid integer input. Please provide valid integers.\n"); return 1; } if (strcmp(argv[2], "and") && strcmp(argv[2], "or") && strcmp(argv[2], "xor")){ printf("Invalid bitwise operation. Valid operations are 'and', 'or', 'xor'.\n"); return 1; } if (!strcmp(argv[2], "and")) { And(num1, num2); } else if (!strcmp(argv[2], "or")) { Or(num1, num2); } else if (!strcmp(argv[2], "xor")) {</pre>			

```

        Xor(num1, num2);
    }
    return 0;
}

void And(int num1, int num2){
    printf("%d ANDed with %d is %d.\n", num1, num2, num1 & num2);
}

void Or(int num1, int num2){
    printf("%d ORed with %d is %d.\n", num1, num2, num1 | num2);
}

void Xor(int num1, int num2){
    printf("%d XORed with %d is %d.\n", num1, num2, num1 ^ num2);
}

```

Program output:

```

PS D:\code\codeInC> gcc -o D:\code\codeInC\test_25_05_26\bitwise D:\code\codeInC\test_25_05_26\bitwise.c
PS D:\code\codeInC> D:\code\codeInC\test_25_05_26\bitwise 12 and 10
12 ANDed with 10 is 8.
PS D:\code\codeInC> D:\code\codeInC\test_25_05_26\bitwise 12 or 10
12 ORed with 10 is 14.
PS D:\code\codeInC> D:\code\codeInC\test_25_05_26\bitwise 12 xor 10
12 XORed with 10 is 6.
PS D:\code\codeInC> D:\code\codeInC\test_25_05_26\bitwise 12 add 10
Invalid bitwise operation. Valid operations are 'and', 'or', 'xor'.
PS D:\code\codeInC> D:\code\codeInC\test_25_05_26\bitwise a or 5
Invalid integer input. Please provide valid integers

```

Analysis:

1. Key Challenges & Solutions

- Understand the usage of command line and the bitwise operator.
- Use the function `sscanf()` to perform the transformation from the type `char` to `int` in the buffer.

2. Result Validation

- Check whether the count of arguments input by command line is valid by the variable `argc`.
- Use the function `sscanf()` and `strcmp()` to respectively check the validity of the integers and the operation.

3. Improvement Suggestions

- May create a function to replace the function `sscanf()` since I forgot it at the beginning but it is not necessary.
- Optimize the input of command line to repeat the input if error.

Conclusion:

In this experiment, we need to learn about the operations of bitwise and their corresponding operators. And we also need to master the usage of command line to input and corresponding format of the `main()` function.